Magnitude & Direction

Exam code: 9MA0

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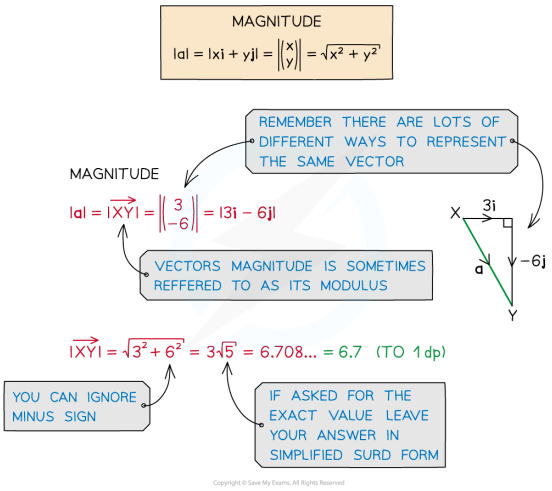
Help me understand

Magnitude & direction

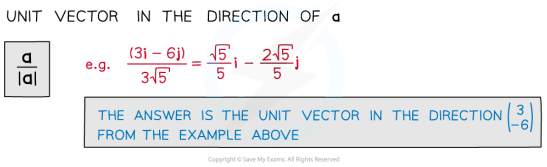
What is the magnitude of a vector?

- The magnitude of a vector is simply its size
- It also tells us the distance between two points
- You can find the magnitude of a vector using Pythagoras' theorem
- The magnitude of a vector **a** is written |**a**| when typed (or |a| when handwritten)



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

So to find the unit vector of a given vector, divide by its magnitude

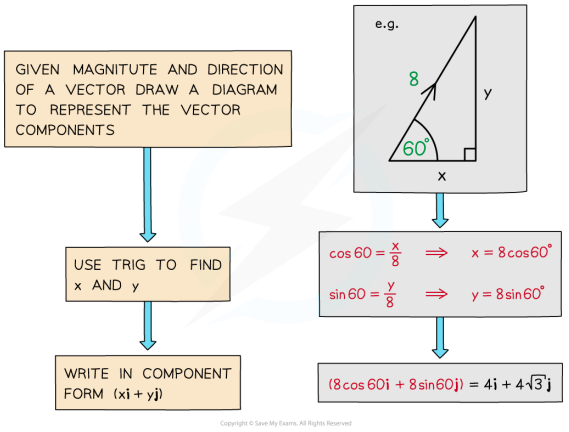


What is the direction of a vector?

- Vectors have opposite direction if they are the same size but opposite signs
  - e.g. if **a** or $\overrightarrow{BC} = \begin{pmatrix} -3 \\ 8 \end{pmatrix}$ then $-\mathbf{a}$ or $\overrightarrow{CB} = \begin{pmatrix} 3 \\ -8 \end{pmatrix}$
- The direction of a vector is what makes it more than just a scalar
  - Eg. two objects with velocities of 7 m/s and -7 m/s are travelling at the **same speed** but in **opposite directions**
- Two vectors are **parallel** if and only if one is a **scalar multiple** of the other
- For real-life contexts such as mechanics, direction can be calculated from a given vector using **trigonometry** (see Right-Angled Triangles)
- It is usually calculated **anticlockwise** from the **positive x-axis** (unless otherwise stated eg. a bearing)

How do I write a vector in component form?

- We have already seen that vectors can be written in different forms
- Component form means writing a vector in terms of **i** and **j components**
- Given the **magnitude** and **direction** of a vector you can work out its components and vice versa



Examiner Tips and Tricks

- Diagrams can help, especially when working out direction – if there isn't a diagram, draw one.
- Remember, resolving a vector just means writing it in component form.

Worked Example

A vector $\overrightarrow{XY} = \begin{pmatrix} -9 \\ 6 \end{pmatrix}$.

a) Find $\left|\overrightarrow{XY}\right|$, giving your answer to 3 significant figures.

b) Write the exact value of the unit vector in the direction of XY, giving your answer in simplified surd form.

a) $|\overrightarrow{XY}| = \sqrt{9^2 + 6^2}$

$= 3\sqrt{13}$

$= 10.8$ (3 sf)

$|\overrightarrow{XY}|$ MEANS FIND THE MAGNITUDE USING PYTHAGORAS

b) USING $\overrightarrow{XY} = \begin{pmatrix} -9 \\ 6 \end{pmatrix}$

$= -9\mathbf{i} + 6\mathbf{j}$

WRITE VECTOR USING i, j NOTATION

AND $|\overrightarrow{XY}| = 3\sqrt{13}$

$\therefore \dfrac{(-9\mathbf{i} + 6\mathbf{j})}{3\sqrt{13}} = \dfrac{-9}{3\sqrt{13}}\mathbf{i} + \dfrac{6}{3\sqrt{13}}\mathbf{j}$

UNIT VECTOR IN GIVEN DIRECTION $= \dfrac{\mathbf{a}}{|\mathbf{a}|}$

$= \dfrac{-3}{\sqrt{13}}\mathbf{i} + \dfrac{2}{\sqrt{13}}\mathbf{j}$

SIMPLIFYING SURD FORM AS MUCH AS POSSIBLE

$= \dfrac{-3\sqrt{13}}{13}\mathbf{i} + \dfrac{2\sqrt{13}}{13}\mathbf{j}$

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